

PA3 – NFAs

(soluções)

1. The difference between a Deterministic Finite Automata (DFA) and a Non-Deterministic Finite Automata (NFA) is:

- ☒ a NFA can have several transitions for the same input symbol, while a DFA can have only one.
 - ☐ a NFA can have infinite states while a DFA can have only a finite number of states.
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2. While processing an input, a NFA can be in more than one state at a time:

- ☒ True
 - ☐ False
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3. There can be no DFA and NFA which accept the same language.

- ☐ True
 - ☒ False
-

4. A dead state can sometimes also be an accepting state.

- ☐ True
 - ☒ False
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5. There exists an algorithm that, for every NFA, returns a DFA which accepts the same language.

- ☒ True
 - ☐ False
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6. NFA cannot have multiple accept states.

☐ True

☒ False

7. What's the lowest number of states possible for a NFA, which accepts a language composed solely of 0's, repeated n times, such that n is a multiple of 3 and n can be zero. 

☐ 2 states

☒ 3 states

☐ 4 states

8. Considering the NFA referred in the previous question, its complement accepts the following strings: "0", "00", and "0000".

☒ True

☐ False

9. Considering a NFA which accepts a language composed of alternating 0's and 1's, but always ending in 0 and any 1 can be replaced with a 0 (accepts "0", "010", "000", "10", "00", "1010", "1000", "0010"), how many states has its corresponding DFA?

☐ 4 states

☒ 5 states

☐ 6 states

☐ 7 states
